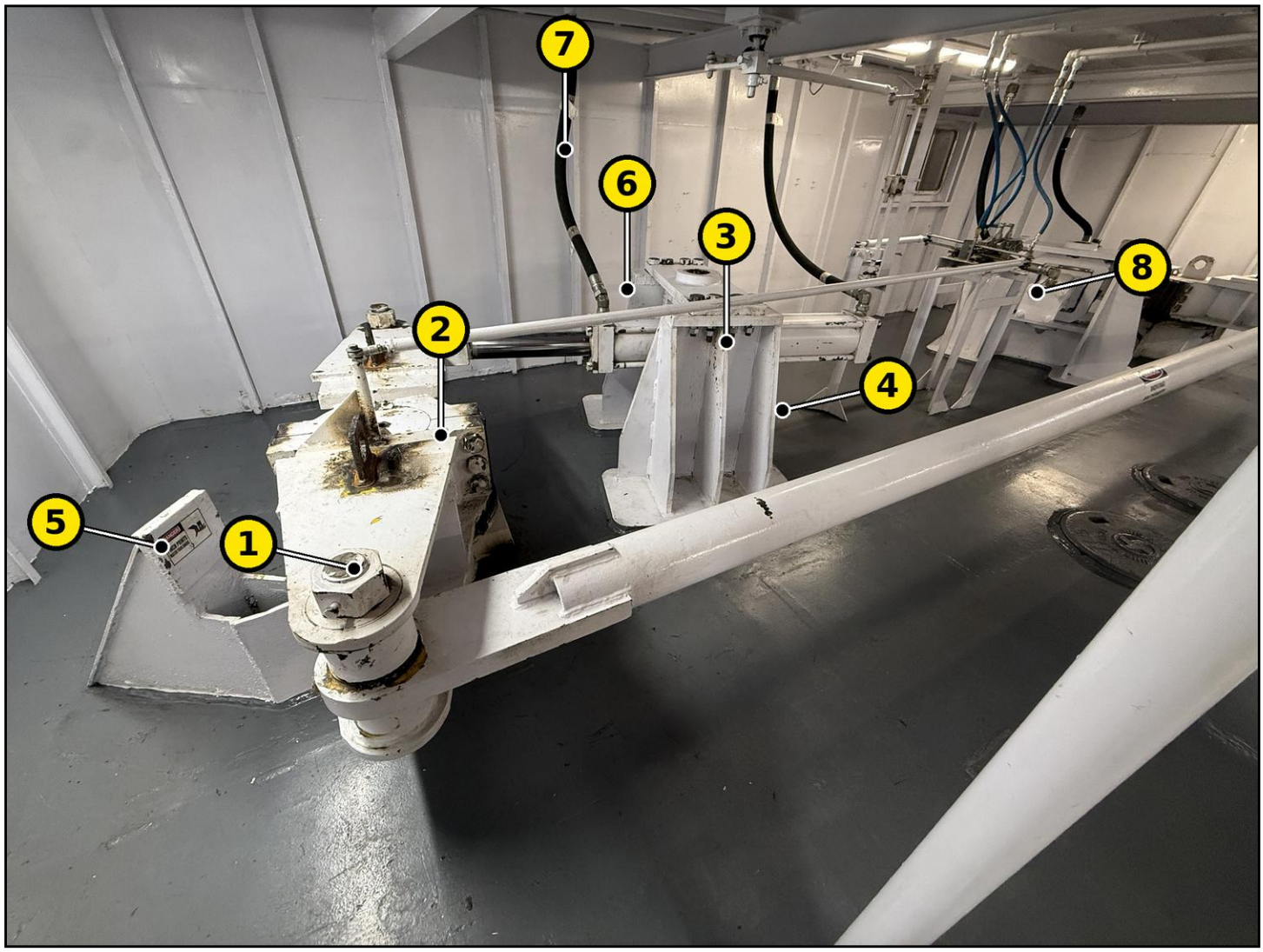
1. STEERING RUDDER GEAR

Steering rudders sit BEHIND the propellers. They steer the boat when the engines are going ahead.



1 Rudder stock & nut
Top of the post the rudder hangs on. It comes up through the deck here. The big nut holds the tiller arm on it.

2 Tiller arm
The heavy arm bolted to the stock. Push the arm, the rudder turns. Everything pushes on this.

3 Steering ram
The hydraulic cylinder on the pedestal. Oil pushes the shiny rod in or out, and the rod swings the tiller arm.

4 Tie bar
The long tube. It links this tiller arm to the other steering rudder's tiller so both rudders move together.

5 Rudder stop
The steel block the tiller hits at hard over. DANGER sticker is there for a reason - never put a hand between the arm and this block.

6 Feedback rod
The thin rod on top. It tells the control valve and the pilothouse gauge where the rudder really is.

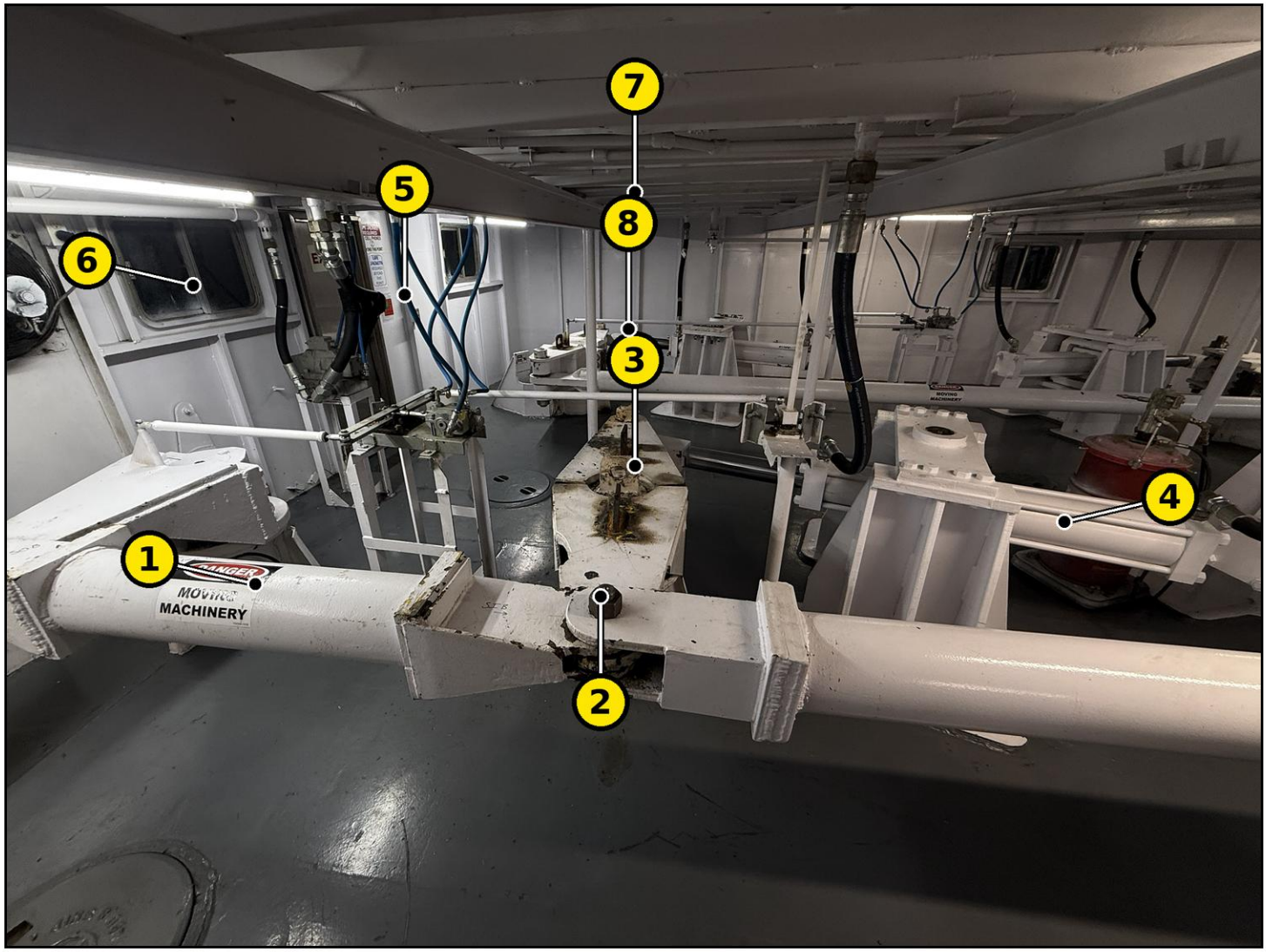
7 Hydraulic hoses
Oil from the steering pumps up in the engine room comes down these hoses to the ram. Over 1,000 psi.

8 Other side
Same setup again for the other steering rudder, on the far side of the room.

WHAT TO LOOK FOR - on rounds: oil under the ram or on the rod, keeper pins on every pin, loose bolts where the arm meets the stock, water weeping where the stock comes through the deck.

2. FLANKING RUDDER GEAR

Flanking rudders sit IN FRONT of the propellers. They steer the boat when the engines are backing. Four of them, two per wheel.



1 Tie bars
The long tubes marked DANGER. They link the flanking rudders so a pair moves together, like a tie rod on a truck's front wheels.

2 Tie bar pin
Where the tie bars pin to the tiller arm. Nut on top, keeper underneath. A pin that walks out means a rudder that's no longer connected.

3 Flanking rudder stock & tiller
The post the flanking rudder hangs on, with its tiller arm. Keep the grease on it - it is a bearing, not decoration.

4 Flanking rams
Hydraulic cylinders on the pedestals that swing the flanking tillers. Their own circuit, worked by the FLANKING lever in the pilothouse.

5 Control valve
Small valve block on the stand with the hoses. The pilot's lever moves this valve, and the valve sends the oil to the rams.

6 Feedback linkage
The thin rods. They move with the tiller and bring the valve back to neutral when the rudder reaches the angle the pilot asked for.

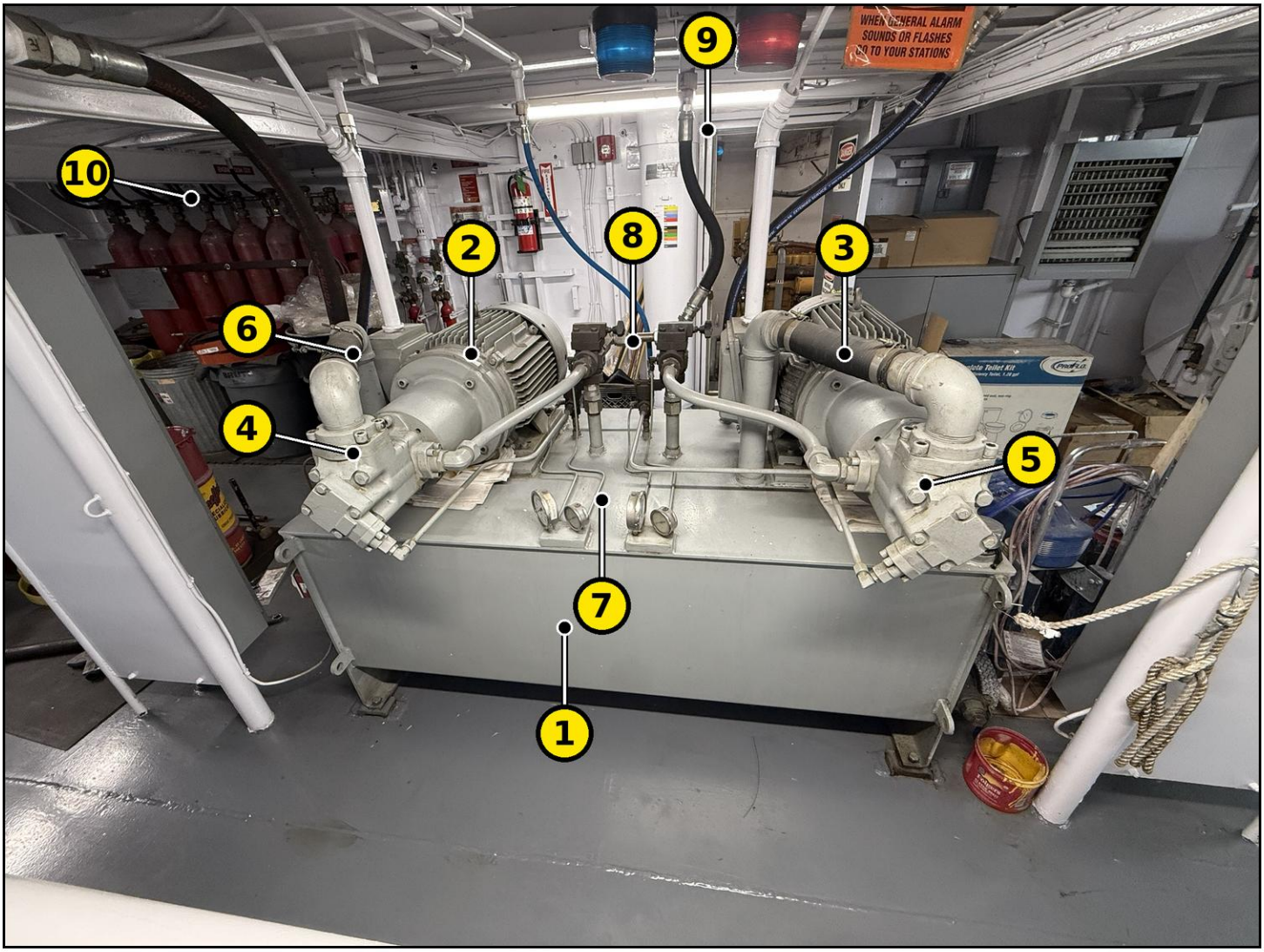
7 Hoses from the pumps
High-pressure oil from the steering pumps. Never feel for a leak with your hand.

8 Steering gear
That's the steering rudder gear from page 1, seen from the other end of the room.

WHAT TO LOOK FOR - on rounds: play in the tie bar ends (pumps OFF and tagged), keepers on every pin, oil on the deck, hoses rubbing on anything.

3. STEERING PUMPS & TANK (engine room)

This is the heart of the system. Two pumps, one tank. One pump runs, the other is the spare. We swap them every turnaround.

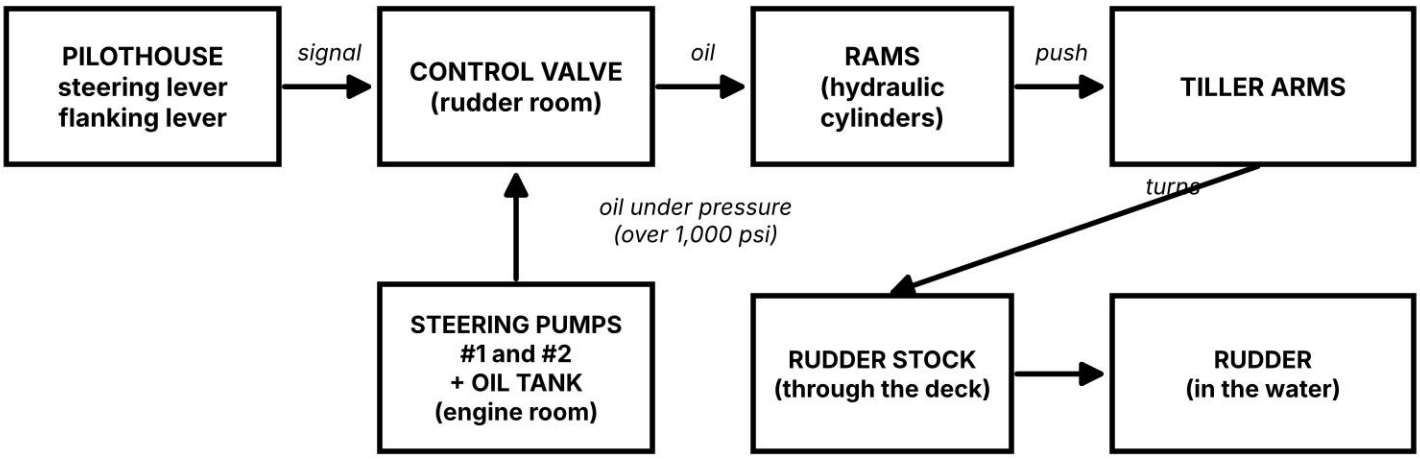


1. **Hydraulic oil tank**
   The reservoir. All the steering oil lives here. Check the level on the sight glass every round - if it's dropping, the oil is leaking out somewhere.
2. **Motor - pump #1**
   Electric motor that drives pump #1.
3. **Motor - pump #2**
   Electric motor that drives pump #2. Only one runs at a time.
4. **Pump #1**
   The pump itself, bolted to the motor. Pulls oil from the tank and pushes it to the rudder room.
5. **Pump #2**
   The spare. Swapped in every turnaround so both wear evenly and the spare is proved to work.
6. **Suction elbow**
   The big elbow is oil coming OUT of the tank into the pump.
7. **Pressure gauges**
   Tell you the running pump is making pressure. Learn what normal reads and report anything lower.
8. **Valves**
   Relief and control valves on top of the tank. Not yours to adjust - know what they are and leave them alone.
9. **Lines to the rudder room**
   The big hoses going up carry oil aft to the rams and bring it back.
10. **NOT steering**
    Red bottles are the CO2 fire system for the engine room. Different system, different training.

WHAT TO LOOK FOR - on rounds: oil level, oil on the deck, pump noise or heat, gauge reading, a tripped breaker. A pump that won't start gets reported right then.

4. THE BIG PICTURE

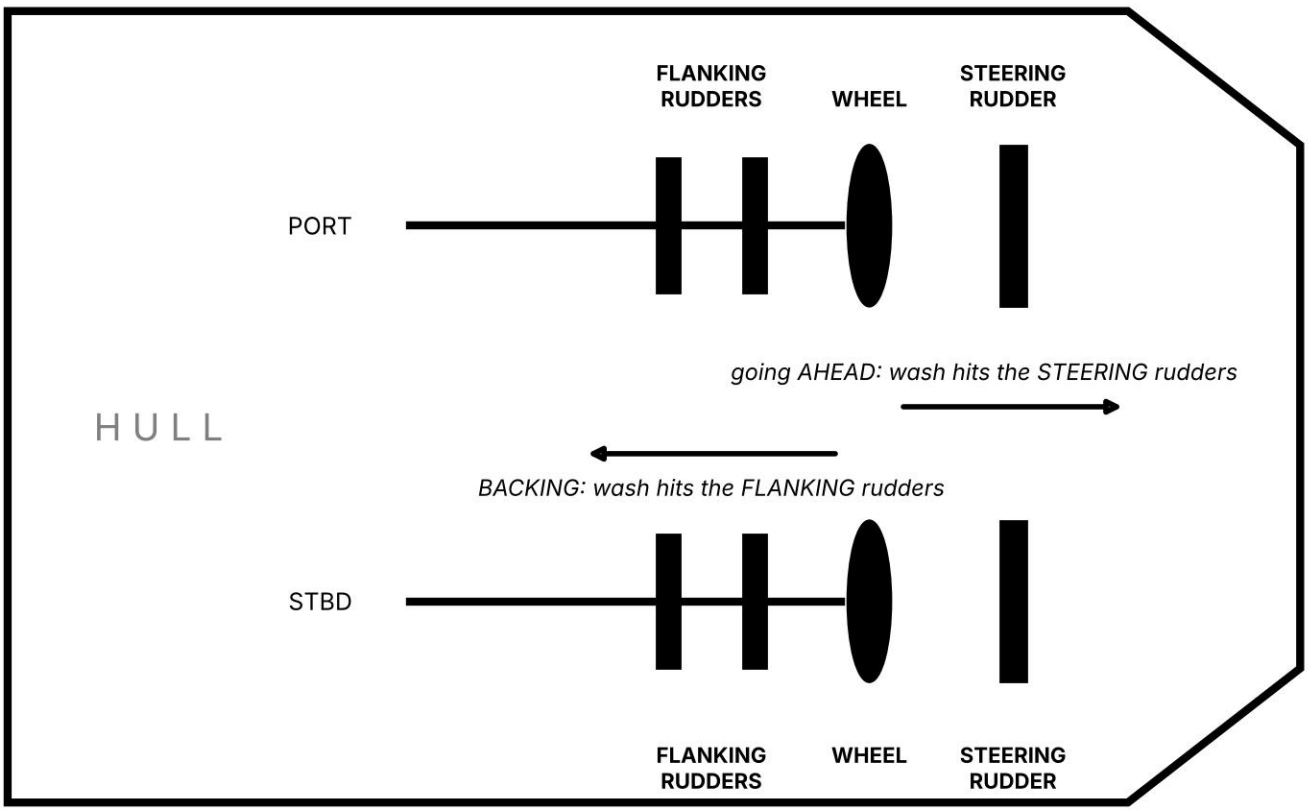
HOW THE PIECES CONNECT



FEEDBACK ROD - as the tiller moves, a thin rod moves the valve back to neutral and sends the true rudder angle to the pilothouse gauge.

RUDDER STOP - a steel block that stops the tiller at hard over. The biggest pinch point in the room.

WHERE THE RUDDERS SIT (looking down on the stern - bow is to the LEFT)



Steering rudders steer going ahead. Flanking rudders steer while backing - that's how a pilot flanks a tow down through a bend.